

Challenge Assignment 2: Social network analysis and modeling

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Exercise 1

Data preparation

In addition to the Facebook network you have used in the Learning-based exercise, you will use the Highschool network here.

Download the “Highschool_network_att.csv”, “Highschool_network_att.csv”, and “LargeFacebook_network_att.csv”, and “LargeFacebook_network_edge.csv” from BrightSpace. Import the .csv data into R and build the two networks.

```
library(igraph)

##
## Attaching package: 'igraph'

## The following objects are masked from 'package:stats':
##
##      decompose, spectrum

## The following object is masked from 'package:base':
##
##      union
```

```
library(RColorBrewer)
library(visNetwork)
```

```
# Highschool network
Highschool_edge <- read.csv("data/Highschool_network_edge.csv", header = FALSE)
Highschool_att <- read.csv("data/Highschool_network_att.csv", header = TRUE)

Highschool_nodes <- data.frame(
  name = as.character(Highschool_att$NodeID),
  gender = as.character(Highschool_att$Gender),
  hall = as.character(Highschool_att$Hall)
)

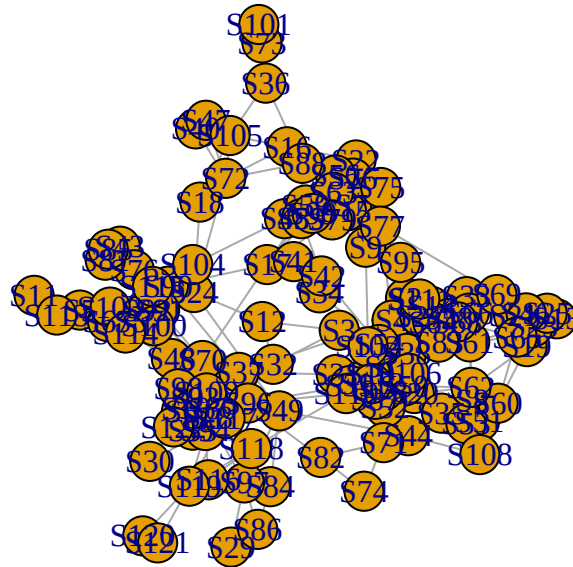
Highschool_edges <- data.frame(
  from = as.character(Highschool_edge[,1]),
  to = as.character(Highschool_edge[,2])
)

Highschool <- graph_from_data_frame(
  Highschool_edges,
  directed = FALSE,
  vertices = Highschool_nodes
)

summary(Highschool)
```

```
## IGRAPH 8547102 UN-- 122 396 --
## + attr: name (v/c), gender (v/c), hall (v/c)
```

```
plot(Highschool)
```



```
# Large Facebook network
LargeFacebook_edge <- read.csv("data/LargeFacebook_network_edge.csv", header = FALSE)
LargeFacebook_att <- read.csv("data/LargeFacebook_network_att.csv", header = TRUE)

LargeFacebook_nodes <- data.frame(
  name = as.character(LargeFacebook_att$NodeID)
)

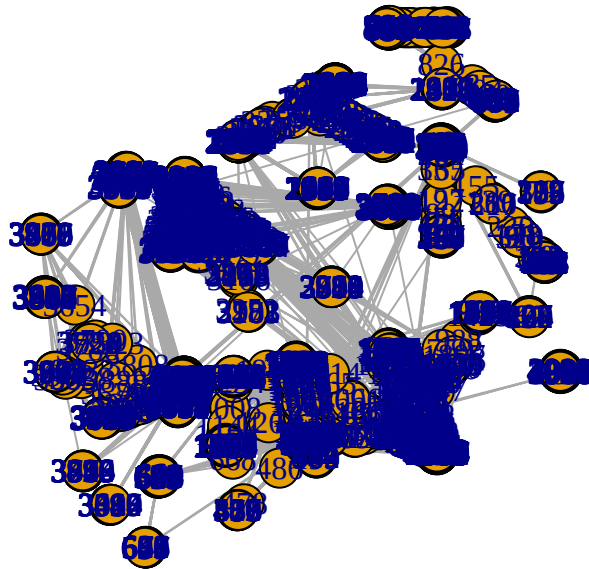
LargeFacebook_edges <- data.frame(
  from = as.character(LargeFacebook_edge[,1]),
  to   = as.character(LargeFacebook_edge[,2])
)

LargeFacebook <- graph_from_data_frame(
  LargeFacebook_edges,
  directed = FALSE,
  vertices = LargeFacebook_nodes
)

summary(LargeFacebook)

## IGRAPH 855f3dc UN-- 4039 88234 --
## + attr: name (v/c)
```

```
plot(LargeFacebook)
```



Question 1 (3 points)

- For both the Highschool and Facebook networks, calculate the shortest path lengths between every pair of two nodes. How many percentage of nodes can be reached within 6 path lengths? Does “six degree of separation” apply to each network?
- Study the degree distribution of these two networks, are they similar? Then use degree distribution to explain the degree of separation you answered above.

You can calculate the shortest path lengths between every pair of two nodes using igraph package:

```
# Shortest path matrix Highschool

dist_highschool <- distances(
  Highschool,
  v = V(Highschool),
  to = V(Highschool),
  mode = "all",
  weights = NULL
)
```

```
# Convert to vector and remove self-distances
dist_high_vec <- dist_highschool[dist_highschool != 0]
```

```
# Percentage within 6 steps
mean(dist_high_vec <= 6) * 100
```

```
## [1] 98.56388
```

```
# Shortest path matrix LargeFacebook
```

```
dist_facebook <- distances(
  LargeFacebook,
  v = V(LargeFacebook),
  to = V(LargeFacebook),
  mode = "all",
  weights = NULL
)
```

```
dist_fb_vec <- dist_facebook[dist_facebook != 0]
```

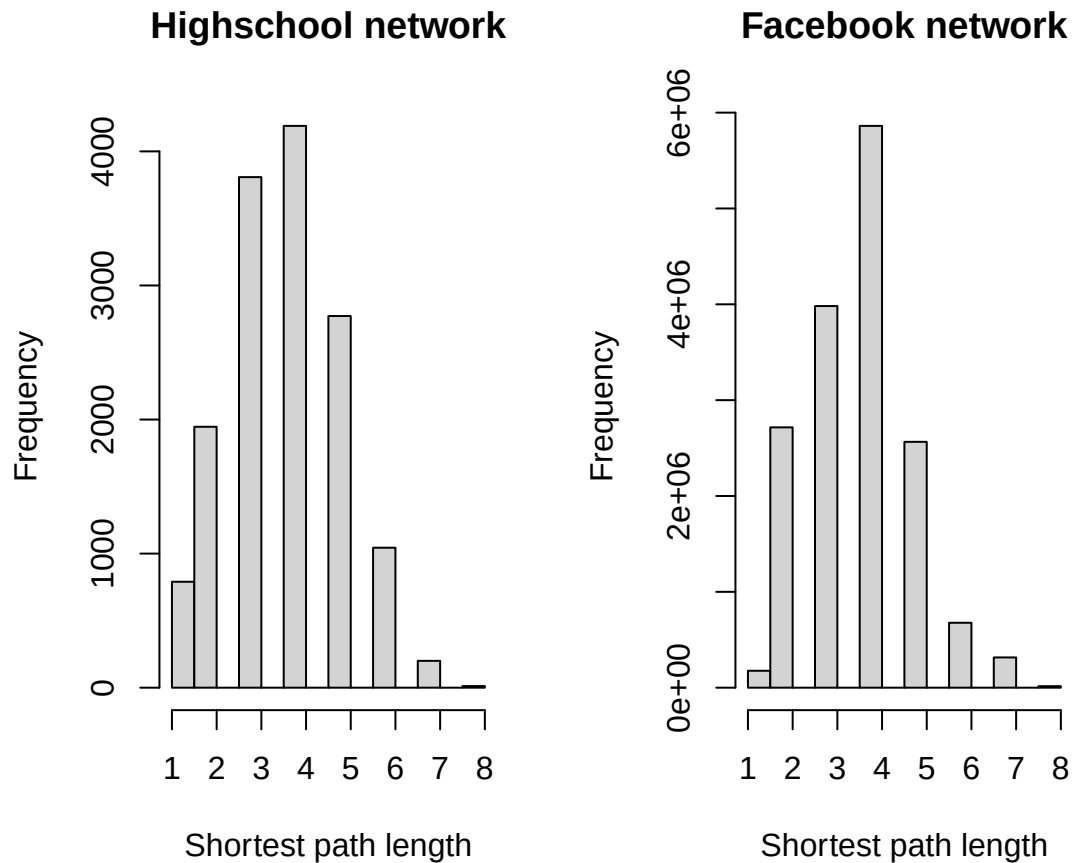
```
mean(dist_fb_vec <= 6) * 100
```

```
## [1] 97.96999
```

```
par(mfrow=c(1,2))
```

```
hist(dist_high_vec,
     breaks=20,
     main="Highschool network",
     xlab="Shortest path length")
```

```
hist(dist_fb_vec,
     breaks=20,
     main="Facebook network",
     xlab="Shortest path length")
```



We computed the proportion of node pairs that are reachable within 6 steps. For the Highschool network this comes out to about 98.6%, and for the Facebook network about 98.0%. In both cases the vast majority of shortest paths lie within 6 steps, so the six degrees of separation idea fits both networks well. In the histograms, most paths in the Highschool network sit around 3–5 steps, with very few beyond 6 and a maximum around 7–8. In the Facebook network, paths tend to be slightly shorter (around 3–4 steps), and almost every pair is reachable within 6 steps.

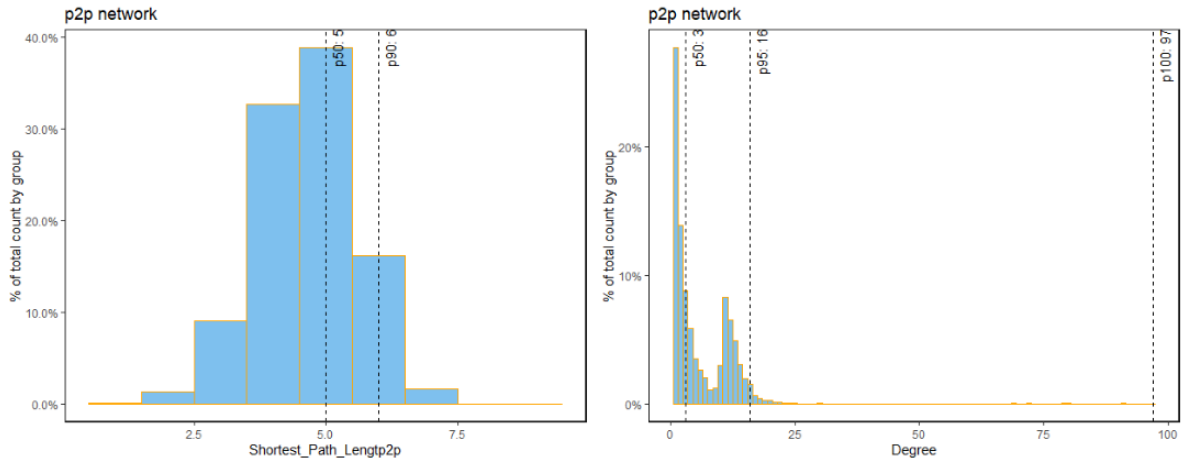
You can consider using a histogram with percentiles to study the shortest path lengths and the degree distribution:

```
deg_high <- degree(Highschool)
deg_fb <- degree(LargeFacebook)

par(mfrow=c(1,2))

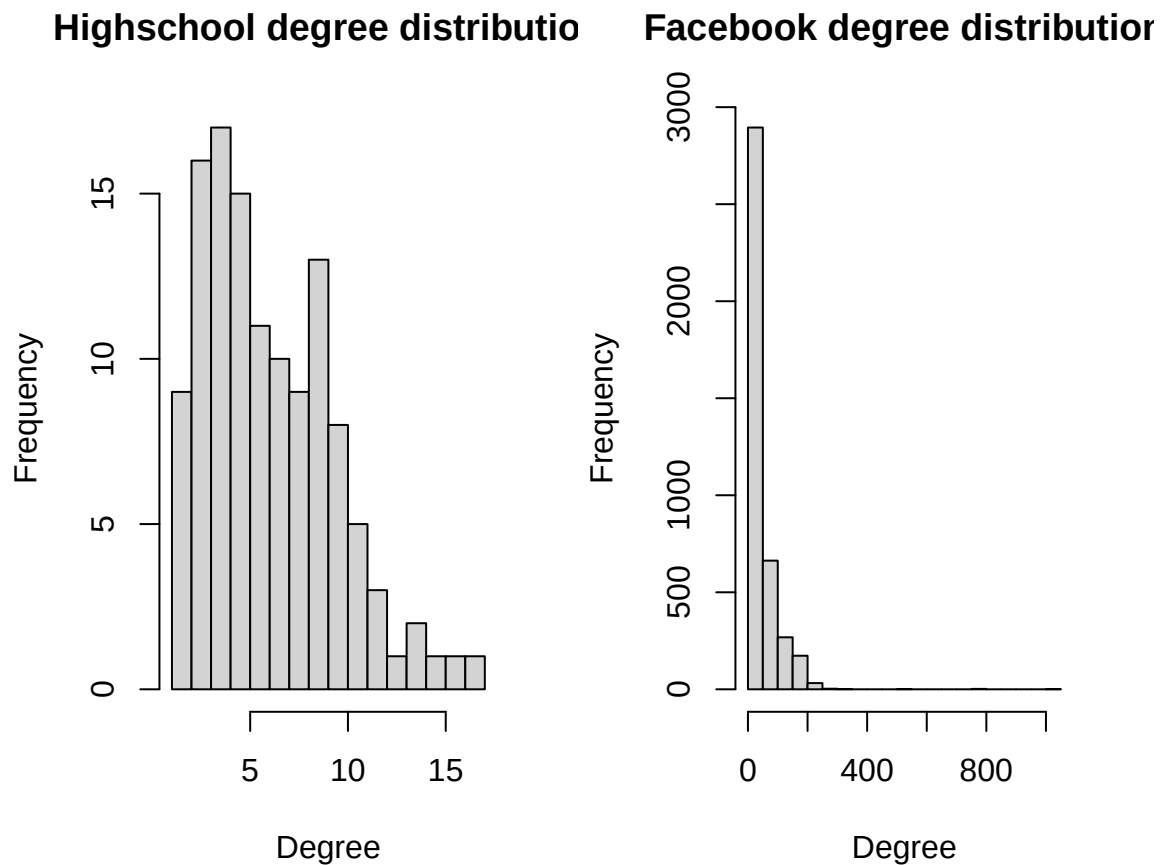
hist(deg_high,
     breaks=20,
     main="Highschool degree distribution",
     xlab="Degree")

hist(deg_fb,
     breaks=20,
     main="Facebook degree distribution",
     xlab="Degree")
```



(Note: the vertical reference line shows the values of shortest path length or the degree at certain percentiles.)

Figure 1: One important stand of community detection algorithm is based on **modularity**, which tries to maximize the difference between the actual number of edges in a community and the expected number of edges in the community. However, optimizing modularity in a network is NP-hard, therefore have to use heuristics.



Degree distribution and shortest path lengths

The two networks have quite different degree distributions. In the Highschool network, the distribution is relatively narrow and moderately right-skewed: most nodes have between about 3 and 8 connections, and the highest degrees are around 15–17. Those nodes have more connections than others but do not dominate the network. The Facebook network, by contrast, is highly right-skewed: most nodes have few connections, while a small number have very high degrees (around 200). Those highly connected nodes act as strong hubs.

This difference helps explain the path lengths we saw above. In the Facebook network, high-degree hubs create shortcuts across the network, so that many otherwise distant pairs are only a few steps apart. In the Highschool network, degrees are more evenly spread and hubs are weaker, so paths depend more on chains of intermediate nodes. Because that network is also small, most pairs are still reachable within six steps; in the larger Facebook network, the presence of hubs does a lot of the work in keeping typical path lengths short.

Question 2 (5 points) *(please submit your code along with your answer)*

- 1) Calculate the modularity of the Highschool network if community is merely identified by a) gender and b) residential hall, respectively.
- 2) Use the Louvain Community Detection to identify communities in the Highschool network. Compare the modularity value produced by the Louvain algorithm to those in 1), and explain the reasons for the differences.

```
# community by gender
genderCommunity <- Highschool_nodes$gender
genderCommunity <- replace(genderCommunity, genderCommunity == "female", 1)
genderCommunity <- replace(genderCommunity, genderCommunity == "male", 2)
genderCommunity <- replace(genderCommunity, genderCommunity == "unknown", 3)
genderCommunity <- as.numeric(genderCommunity)

gender.clustering <- make_clusters(Highschool, membership = genderCommunity)
mod_gender <- modularity(gender.clustering)
```

```
#community by residential hall
hallCommunity <- as.factor(Highschool_nodes$hall)
hallCommunity <- as.numeric(hallCommunity)

hall.clustering <- make_clusters(Highschool, membership = hallCommunity)
mod_hall <- modularity(hall.clustering)
```

```
#Louvain algorithm
Louv <- cluster_louvain(Highschool)
mod_louv <- modularity(Louv)
```

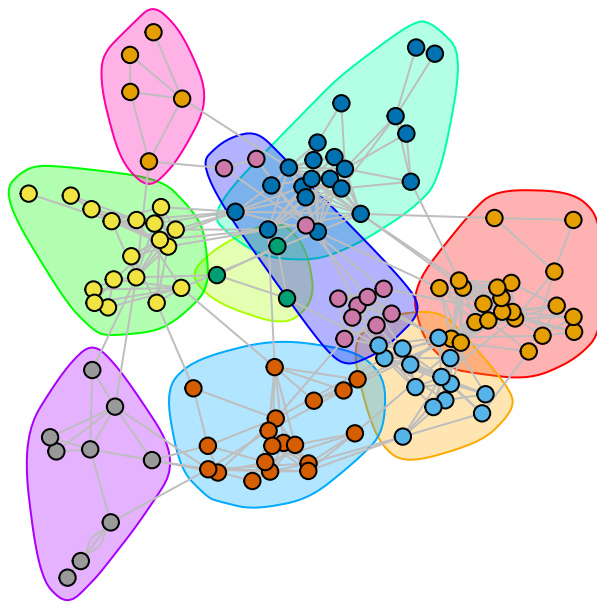
```
data.frame(
  method = c("Gender", "Residential Hall", "Louvain"),
  modularity = c(mod_gender, mod_hall, mod_louv)
)
```

```
##           method  modularity
```

```
## 1          Gender 4.782675e-05
## 2 Residential Hall 1.755911e-01
## 3          Louvain 7.016599e-01
```

```
plot(
  Louv,
  Highschool,
  layout = layout_with_fr(Highschool),
  vertex.size = 6,
  vertex.label = NA,
  edge.color = "gray",
  mark.groups = communities(Louv),
  main = "Louvain communities in the Highschool network"
)
```

Louvain communities in the Highschool network



Answer

We compared three ways of defining communities on the Highschool network: by gender, by residential hall, and using the Louvain algorithm. The modularity values tell a clear story.

When we partition the network by gender, modularity is extremely low (about 0.00005). Friendships in this network are not organised along gender lines; male and female students are often connected to each other

rather than forming separate groups. When we partition by residential hall, modularity is higher (about 0.176). Students are somewhat more likely to be friends with others in the same hall, so hall membership captures part of the structure—but only part, and the value remains moderate.

The Louvain algorithm does much better, with modularity around 0.701. The reason is that gender and hall are fixed attributes we impose on the network, whereas Louvain groups nodes directly from their connection patterns and thus optimizes modularity. The communities it finds therefore align much more closely with how ties are actually distributed. In the plot we see densely connected clusters with relatively few links between them—likely friend circles, shared activities, or overlapping groups that are not neatly summarized by a single attribute like gender or hall. So in short: gender explains almost none of the structure, hall explains some of it, and Louvain recovers the strongest community structure because it uses the topology of the network itself.

Exercise 2

You might realize not every value of p can return you a small-world network that you are looking for. Then a question arises as how can one find the range of p . In the Figure 2 of Watts and Strogatz (1998), it explains how can one decide the range of p by looking at the dynamics between path length and clustering coefficient.

Question 3 (4 points)

- 1) Start with a regular network of size=300, $nei=6$, first reproduce the Figure 2 of Watts and Strogatz (1998). Then provide the range of p which can turn this regular network (size=300, $nei=6$) into a small-world network.
- 2) Do you need to rewired significant amount of connections to make the network small-world-like?
- 3) In the paper of Watts and Strogatz (1998), they pointed out that the value of p has two important implications:
 - “The idealized construction above reveals the key role of short cuts. It suggests that the small-world phenomenon might be common in sparse networks with many vertices, as even a tiny fraction of short cuts would suffice.”
 - “Thus, infectious diseases are predicted to spread much more easily and quickly in a small world; the alarming and less obvious point is how few short cuts are needed to make the world small.”

Use your own words to explain these two implications. For the second implication, connect it with the spread of COVID.